

LESSON 19 / 第十九課

Trees and Underground Utilities: From the Shared Cubic Metre to the Cut Face

樹木與地下管線：從共享立方公尺到切面

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Trees and utilities occupy the same shallow volume. The upper metre of a street section is a shared resource, not two separate drawings. Clearance tables are spatial rules, not root maps. Each trench is one of the four construction injury classes. Absorbing-root loss and structural-root severance tax different accounts. A dual residual-risk sentence is required: service continuity and tree risk. Backfill chemistry and compaction after the trench are part of the injury.

樹木與管線佔據同一淺層體積。街道斷面上層一公尺是共享資源，不是兩張分開的圖。淨距表是空間規則，不是根系圖。每條溝是四種施工傷害類別之一。吸收根喪失與結構根切斷抽不同賬的稅。需要雙重剩餘風險句子：服務連續與樹木風險。開溝後的回填化學與壓實是傷害的一部分。

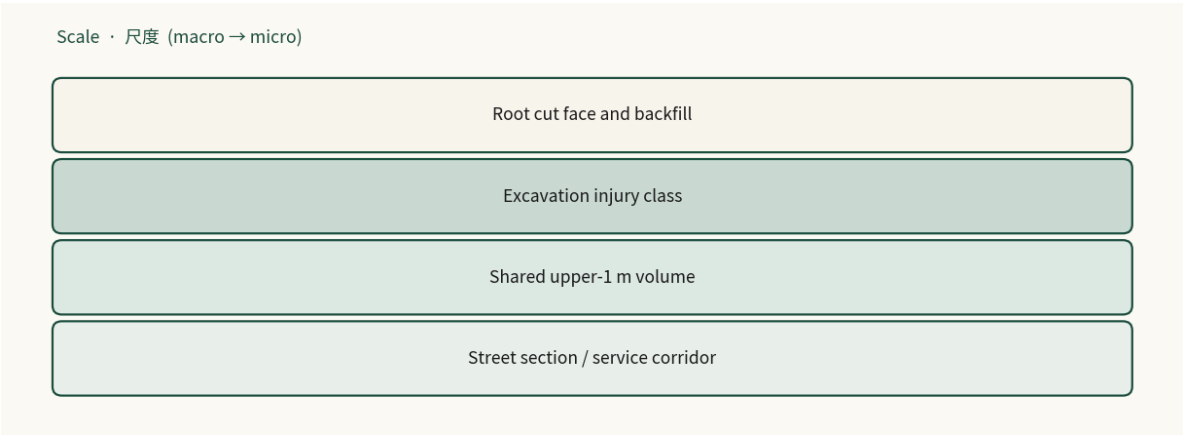


Figure 1. Nested scales from the corridor to the cut face.

圖 1. 由走廊到切面的嵌套尺度。

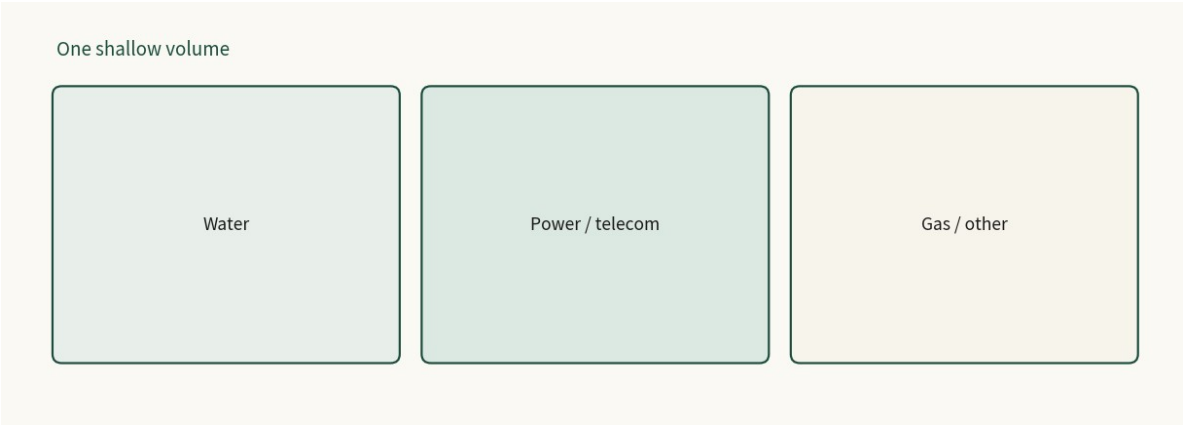


Figure 2. Utilities share the upper metre with roots.

圖2. 管線與根系共享上層一公尺。

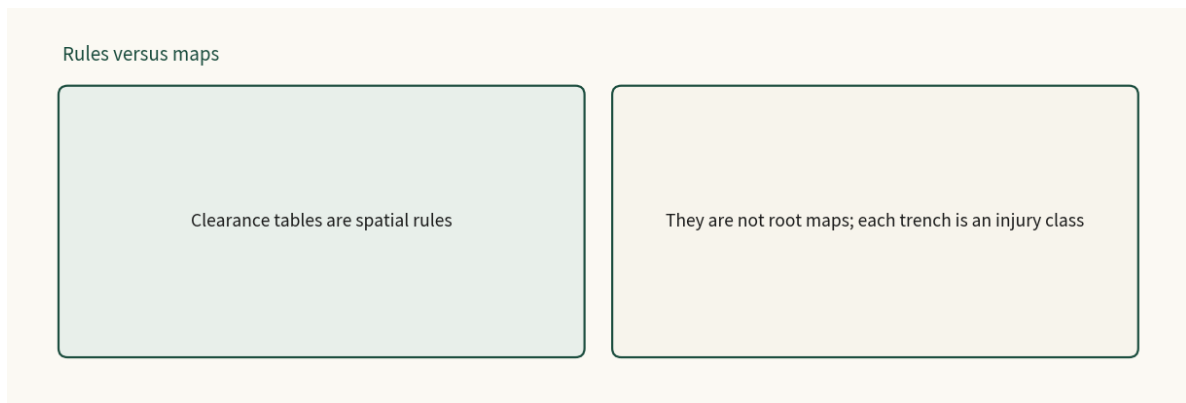


Figure 3. Clearance tables are rules, not photographs of roots.

圖3. 淨距表是規則，不是根系照片。

1. Scope

1. 範圍

The object is published conflict and coexistence of tree roots with underground services.

對象是樹根與地下服務的已發表衝突與共存。

2. Shared volume

2. 共享體積

Absorbing roots of most urban trees occupy the upper soil where trenches also run [1,2]. Treating the two as separate drawings produces surprise at the trench wall. Costello and Jones treat the conflict as a design failure more often than as a mysterious property of a taxon [3].

多數城市樹的吸收根佔據溝槽也在走的上層土 [1,2]。把兩者當成分開的圖，會在溝壁處產生意外。Costello 與 Jones 更常把衝突當成設計失敗，而不是某一分類群的神秘屬性 [3]。

3. Injury class and dual sentence

3. 傷害類別與雙重句子

Each trench is one of the four construction injury classes. Structural-root cuts change the plate; absorbing-root loss changes the water bill [4]. Service continuity and tree risk are two residual-risk sentences. Backfill that is over-compacted or chemically hostile extends the injury after the pipe is in.

每條溝是四種施工傷害類別之一。結構根切斷改變盤；吸收根喪失改變水帳單 [4]。服務連續與樹木風險是兩句剩餘風險。過度壓實或化學不友善的回填，在管子就位之後延長傷害。

4. Limits

4. 限度

Clearance distances calibrated on temperate open soil transfer poorly under sealed pavement and into stonewall root mats. Where a root map exists, the map beats the table.

在溫帶開敞土壤上率定的淨距，難以轉到封蓋鋪面下與石牆根墊裡。凡根系圖存在之處，圖勝於表。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Trees and utilities occupy:

1. 樹木與管線佔據：

- A. Separate planets.
- A. 分開的行星。
- B. The same shallow volume.
- B. 同一淺層體積。
- C. Only P50.
- C. 只有 P50。
- D. Only Wall 4.
- D. 只有第 4 壁。

2. Clearance tables are:

2. 淨距表是：

- A. Root photographs.
- A. 根系照片。
- B. Spatial rules, not root maps.
- B. 空間規則，不是根系圖。
- C. Vulnerability curves.
- C. 脆弱性曲線。
- D. Registers.
- D. 名錄。

3. Each trench is:

3. 每條溝是：

- A. Free of injury by definition.
- A. 定義上無傷害。
- B. One of the four construction injury classes.
- B. 四種施工傷害類別之一。
- C. A 10-20-30 rule.
- C. 10-20-30 規則。
- D. A flash map.
- D. 閃擊圖。

4. Structural-root cuts and absorbing-root loss:

4. 結構根切斷與吸收根喪失：

- A. Are the same bill.
- A. 是同一帳單。
- B. Tax different accounts.

B. 抽不同賬的稅。

C. Equal occupancy.

C. 等於佔用。

D. Equal SOS.

D. 等於 SOS。

5. The required sentence is:

5. 需要的句子是：

A. One-way, utilities only.

A. 單向，只有管線。

B. Dual: service continuity and tree risk.

B. 雙重：服務連續與樹木風險。

C. Paint only.

C. 只有漆。

D. i-Tree only.

D. 只有 i-Tree。

6. Backfill after the trench:

6. 開溝後的回填：

A. Is a coda outside the injury.

A. 傷害之外的尾聲。

B. Is part of the injury.

B. 是傷害的一部分。

C. Raises P88.

C. 提高 P88。

D. Is Wall 1.

D. 是第 1 壁。

7. Surprise at the trench wall usually means:

7. 溝壁處的意外通常意味：

A. Trees are mystical.

A. 樹是神秘的。

B. The two objects were drawn separately.

B. 兩個對象被分開畫。

C. Choat 2012 failed.

C. Choat 2012 失敗。

D. A palm meristem moved.

D. 棕櫚分生組織移動了。

8. Where a root map exists:

8. 凡根系圖存在：

A. The table remains sovereign.

A. 表仍為主權。

B. The map beats the table.

B. 圖勝於表。

C. t/R is sovereign.

C. t/R 為主權。

D. Paint is sovereign.

D. 漆為主權。

9. Conflict is more often:

9. 衝突更常是：

A. A mysterious taxon property.

A. 神秘的分類群屬性。

B. A design failure of shared volume.

B. 共享體積的設計失敗。

C. A sacred month.

C. 神聖月份。

D. A cable colour.

D. 鋼纜顏色。

10. Sealed pavement above services:

10. 服務上方的封蓋鋪面：

A. Makes temperate clearance tables more accurate.

A. 使溫帶淨距表更準。

B. Makes those tables transfer more poorly.

B. 使那些表更難轉移。

C. Equals 3-30-300.

C. 等於 3-30-300。

D. Equals QTRA.

D. 等於 QTRA。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Day S.D. et al. (2010). urban root ecology. AUF 36: 193-205.

[2] Grabosky J., Bassuk N. shared-volume work.

[3] Costello L.R., Jones K.S. reducing infrastructure and tree conflicts.

[4] Watson G.W. root-severance recovery.

[5] Fite K., Smiley E.T. construction BMPs applied to utility trenches.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。